Big Data Project 2 Implementation

LLM-Powered Dataset Description Generation

Apache Spark and Parallelized Mining Applications

Presented by Team TomCat

Project Proposal

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Problem Summary:

With millions of datasets publicly available on platforms like data lakes and open data portals, effective discovery remains a significant bottleneck. Many datasets suffer from missing or inadequate descriptions, rendering them virtually invisible to search engines. AutoDDG is a framework that leverages algorithmic data profiling and Large Language Models (LLMs) to synthesize content and semantic summaries.

This project answers the question, how effectively can the AutoDDG framework generate high-quality, searchable descriptions for open-domain tabular datasets compared to human-authored descriptions? How can we evaluate the creation of searchable and complete descriptions for datasets that have no prior dataset dictionary? How do we expand this to data released by scientific publications and enable their discovery by search engines so that such data can be found more easily besides domain experts with their associated patterns? As a stretch goal, how do we incorporate multimodality into searchable and complete descriptions generated by AutoDDG when images, video and text are present in research data? How can TabLib be used to expand on and describe tabular data?

How will we solve the problem:

Our two main methodologies are

- Evaluating semantic alignment and searchability using OpenAI models as judges and lexical/embedding metrics.
- Profiling and running pattern data analysis on large dataset collections using Apache Spark to verify that generated descriptions accurately capture underlying column data patterns.

The source of our primary datasets will be from NYC Open Data and Auctus, as specified in the project description. We will employ filters present in the data portals to find representative popular datasets with complete data dictionaries and missing data dictionaries. We plan to use Apache Spark to easily identify patterns present in the complete descriptions to validate AutoDDG descriptions and also use the OpenAI scoring tool present in the framework to compare and contrast searchability and completeness for generated descriptions or if existing descriptions need to be revised for better searchability. We will employ APIs present in both the NYC Open Data and Auctus frameworks to load the data into Spark via HDFS and use SQL operations to identify patterns - this framework will expand when we are working with publicly available datasets found on platforms like *Opportunity Insights* and *Zenodo*. For multimodal datasets, we will also use APIs when available to load data into Spark and identify ways to load images (possibly in base64?) but first tables into the file system to identify patterns and enrich the dataset descriptions and allow them to be found easier on SEO platforms.

Milestones:

Milestone 1: Setup, Baseline Selection, & Environment Prep (Weeks 1–3)

- **Goal:** Establish the technical foundation and dataset scope.
- **Key Tasks:**
    - Clone and configure the AutoDDG framework.
    - Curate dataset samples from **NYC Open Data** and **Auctus** that already contain human-authored descriptions (to serve as ground truth).
    - Initialize the Apache Spark environment for handling larger data files.

Milestone 2: Description Generation & Spark Pattern Analysis (Weeks 4–7)

- **Goal:** Execute the core methodology for generating summaries and analyzing underlying data.
- **Key Tasks:**
    - Run AutoDDG across the target NYC Open Data and Auctus datasets to produce content/semantic summaries and final descriptions.
    - Load large dataset collections into Spark to extract deep column-level statistics, value distributions, and structural patterns.
    - Cross-reference Spark-discovered data patterns against the generated text to ensure descriptive accuracy.
    - Use OpenAI scoring metrics to compare generated dataset descriptions and ground truth descriptions for searchability and completeness.

Milestone 3: Evaluation & Stretch Exploration (Weeks 8–10)

- **Goal:** Benchmark description quality and explore optional stretch goals.
- **Key Tasks:**
    - Implement the OpenAI-as-a-judge scoring mechanism alongside lexical metrics to evaluate how well new descriptions match original human descriptions in descriptive ability and searchability.
    - Use another agent-as-a-judge framework to qualitatively judge searchability and completeness for AutoDDG descriptions and use the OpenAI scoring tool to quantitatively judge searchability and completeness.
    - Curate dataset dictionaries for datasets with no data dictionary on OpenNYC Data and Auctus using Spark-mined big data mining.
    - (*Stretch*) Expand the methodology to research-associated datasets or preliminary multimodal data if time permits.

Milestone 4: Final Report & Wrap-Up (Weeks 11–12 / Deadline: December 9)

- **Goal:** Synthesize all findings and compile the final report.
- **Key Tasks:**
    - Draft the Problem Statement detailing the visibility issue in data portals.
    - Outline the Approach (AutoDDG, OpenAI evaluation, Spark patterns, and datasets).
    - Document Key Findings (quantitative scores, searchability impact, and Spark pattern insights).
    - Write the Reflection covering API limits, prompt sensitivities, and system limitations.